Bani sueif national university

Faculty of computers and Ai

Banking System Project

Under the supervision of : *DR. Hayam Mohmed Sayed*

Submitted by : Abdelrahman Nagm-eldin → G1 SEC 3

INTRODUCTION:

1. **This report presents an overview of the requirements and implementation stages of the Database Systems course project.**

2. **A fully functional database was created, and the tables were properly designed and structured.**

3. **Records were successfully inserted into the database. In addition, Structured Query Language (SQL) commands studied during the course were applied.**

4. **A user interface was also developed to facilitate interaction with the database, and it was integrated with the database using the Python programming language.**

REQUIREMENTS:

1. **ERD & SCHEMA**
2. **Each table contains a constraint**
3. **All the SQL commands that were previously studied.**
4. **GUI contains (insert , update , delete)**

NOTES:

i) In ERD :-

- **Computed attribute "NUMBER OF EMPLOYEES " retrieved by**

```
SELECT
   d.name AS department_name,
   COUNT(e.employee_id) AS total_employees
FROM
   department d
LEFT JOIN
   employee e ON d.department_code = e.department_code_FK
GROUP BY
   d.department_code, d.name;
```

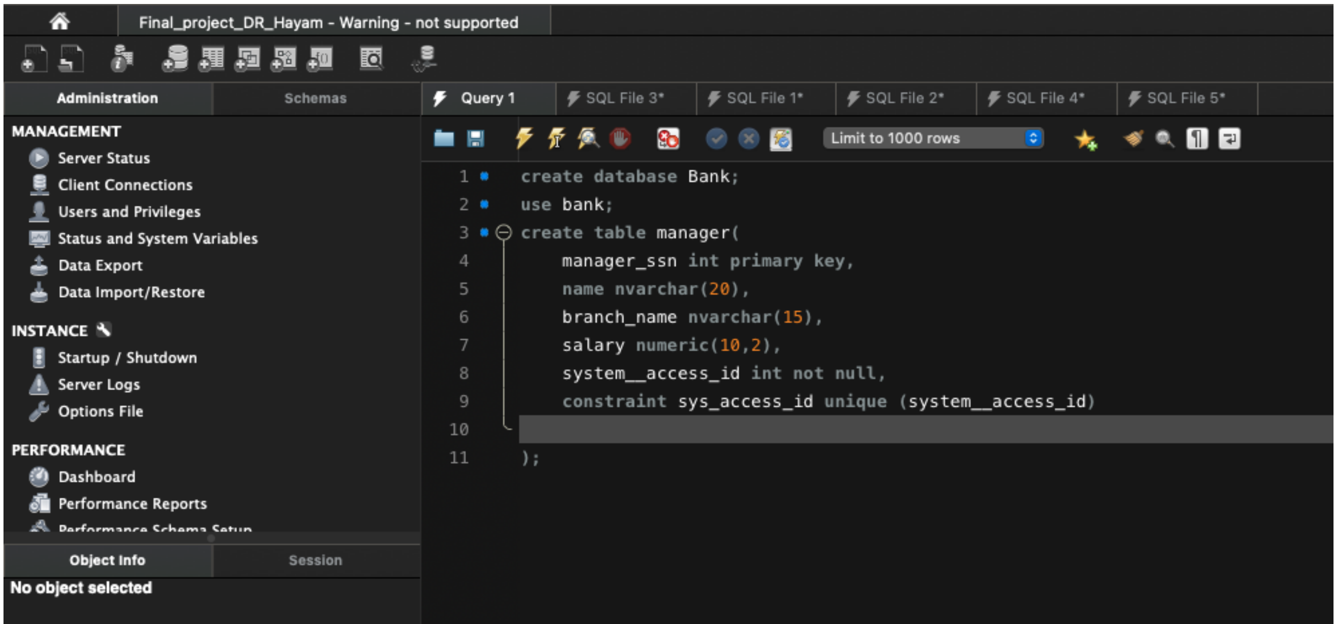
ii) In schema

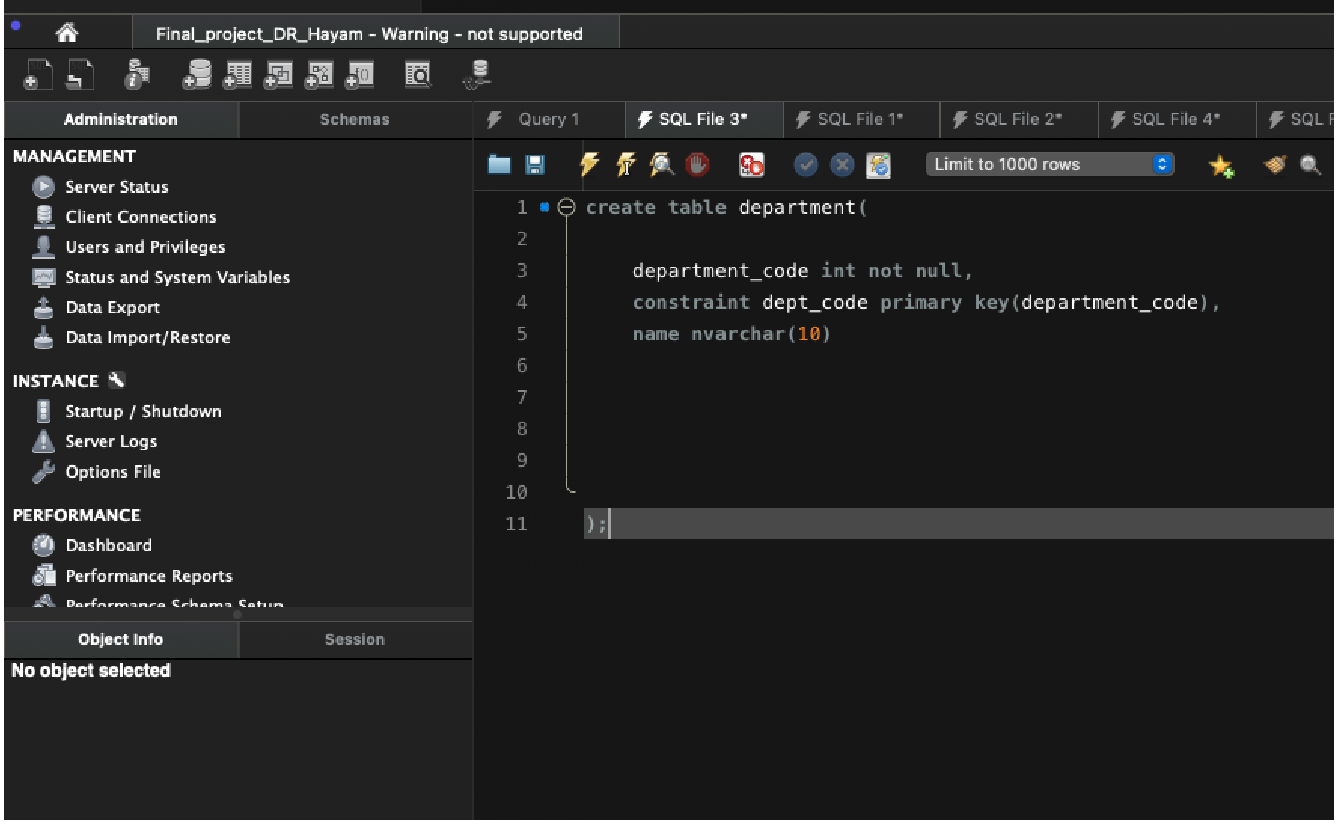
- **We add 2 tables because the multivalue attribute "PHONE" and the relationship many to many "DEAL WITH"**
- **P.K went from 1 to many**
- **P.K went from partial participation to total participation**

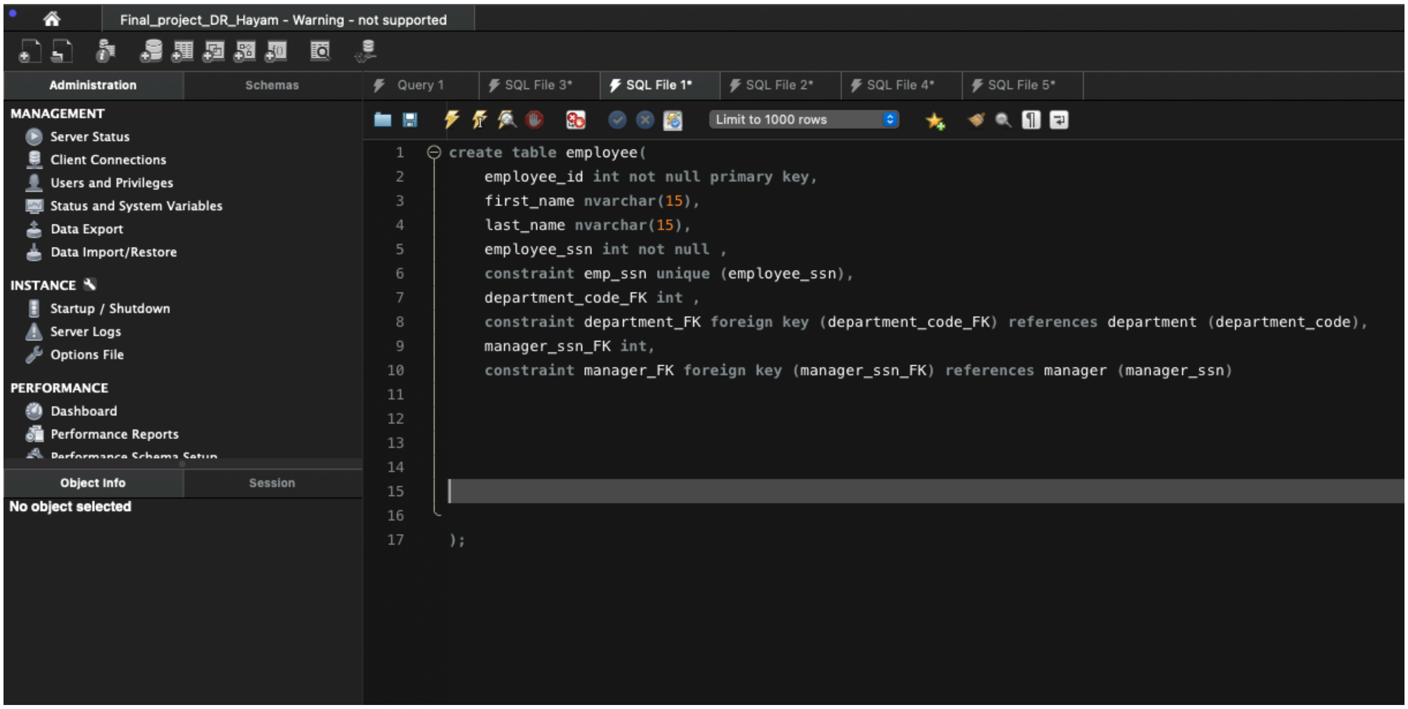
SQL commands:-

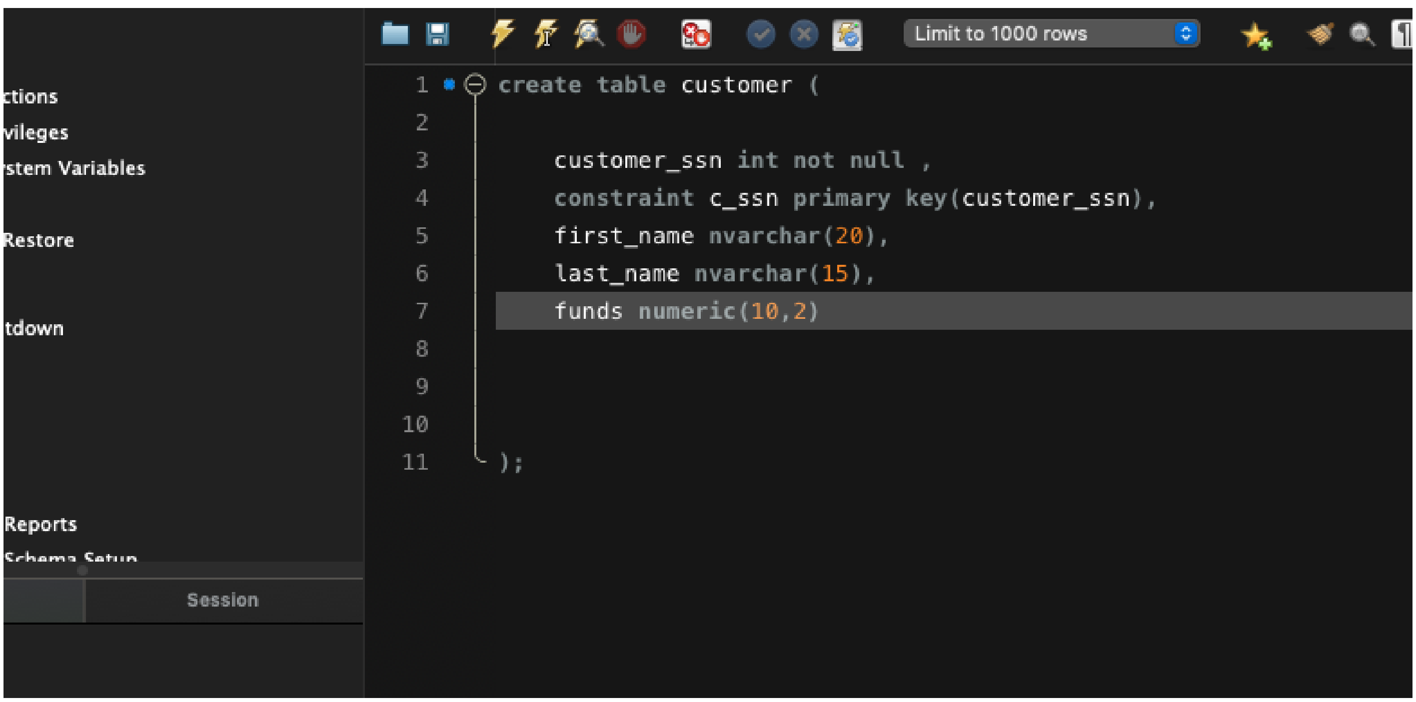
DDL:-

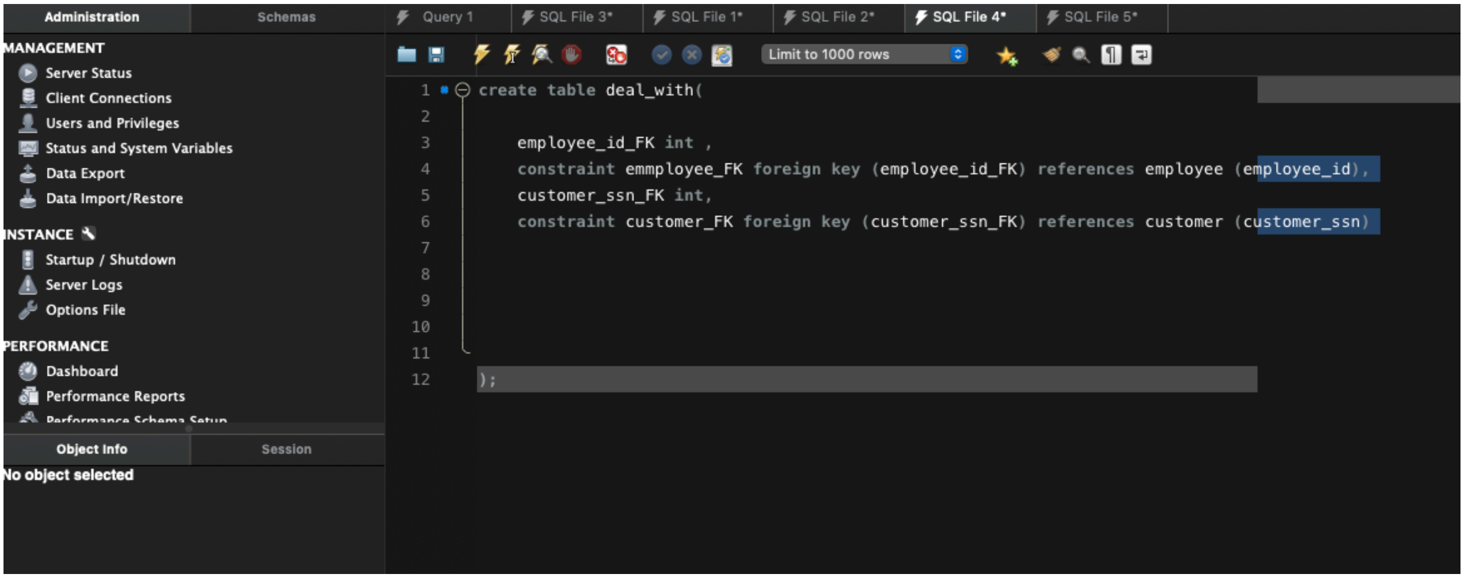
- **CREATE**

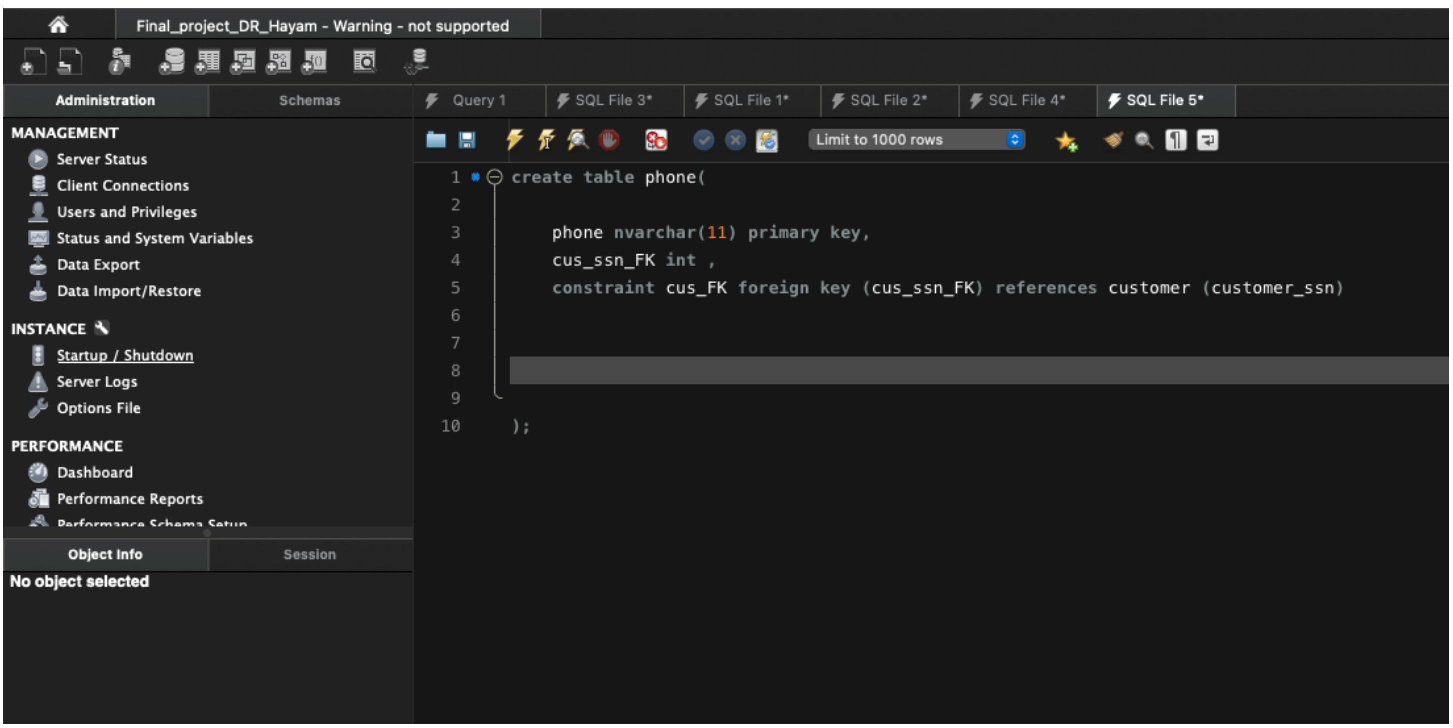




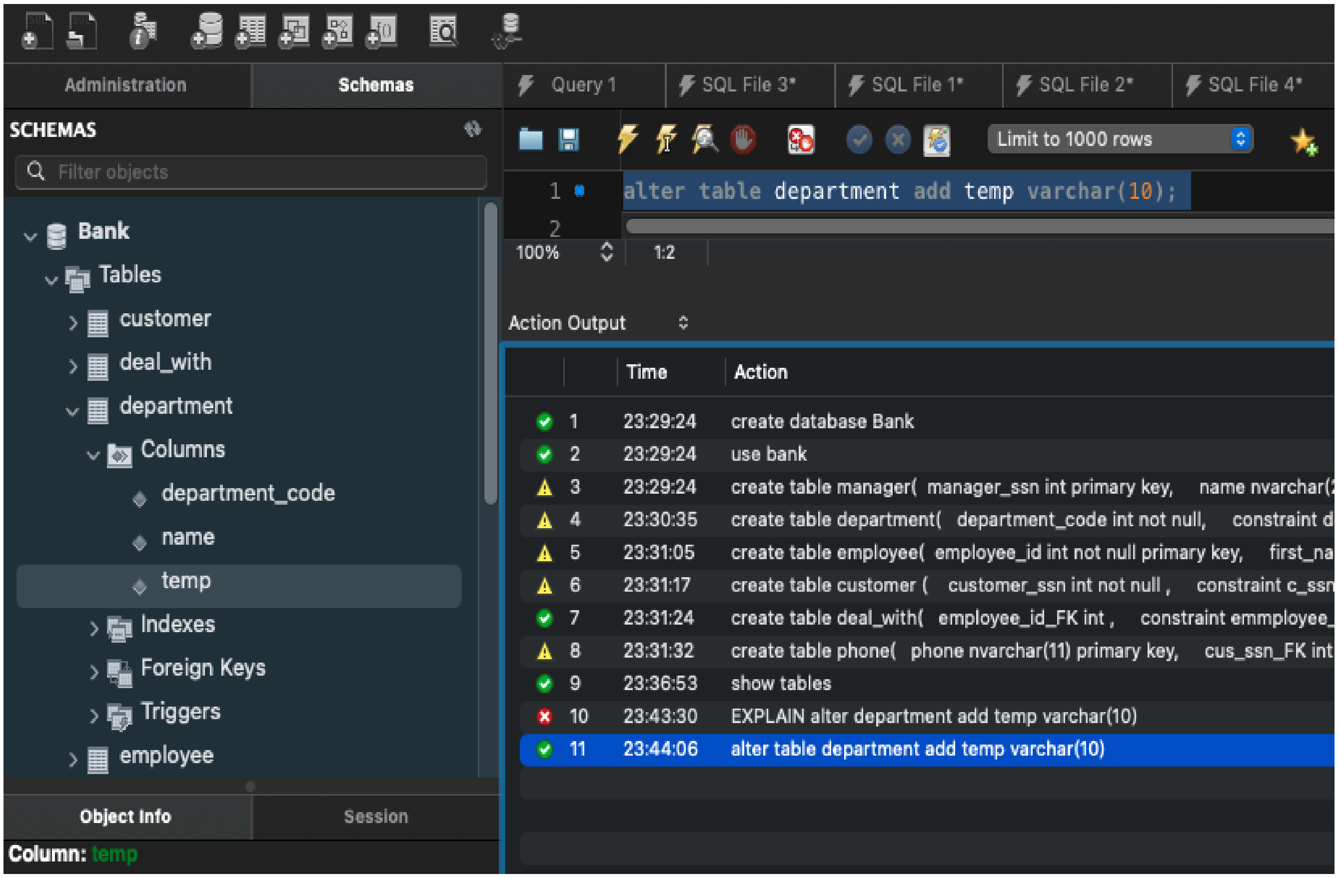

Final_project_DR_Hayam - Warning - not supported
Administration
Schemas
Query 1
SQL File 3*
SQL File 1*
SQL File 2*
SQL File 4*
SQL File 5*
MANAGEMENT
Server Status
Client Connections
Users and Privileges
Status and System Variables
Data Export
Data Import/Restore
INSTANCE
Startup / Shutdown
Server Logs
Options File
PERFORMANCE
Dashboard
Performance Reports
Object Info
Session
No object selected
Limit to 1000 rows
create table employee(
employee_id int not null primary key,
first_name nvarchar(15),
last_name nvarchar(15),
employee_ssn int not null ,
constraint emp_ssn unique (employee_ssn),
department_code_FK int ,
constraint department_FK foreign key (department_code_FK) references department (department_code),
manager_ssn_FK int,
constraint manager_FK foreign key (manager_ssn_FK) references manager (manager_ssn)
);


Limit to 1000 rows
ctions
vileges
stem Variables
Restore
tdown
Reports
Session
create table customer (
customer_ssn int not null ,
constraint c_ssn primary key(customer_ssn),
first_name nvarchar(20),
last_name nvarchar(15),
funds numeric(10,2)
);

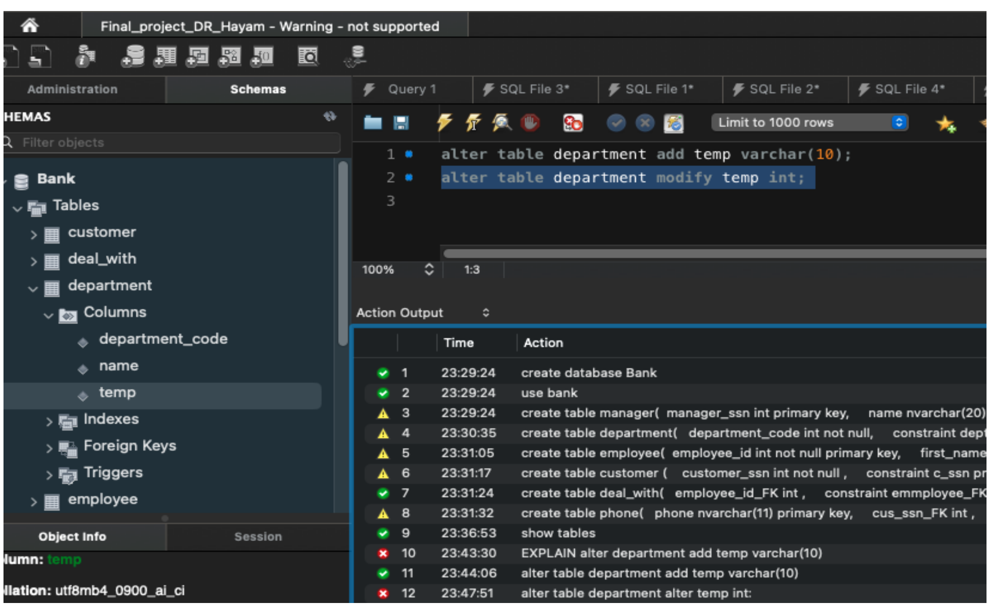

Administration
Schemas
Query 1
SQL File 3*
SQL File 1*
SQL File 2*
SQL File 4*
SQL File 5*
MANAGEMENT
Server Status
Client Connections
Users and Privileges
Status and System Variables
Data Export
Data Import/Restore
INSTANCE
Startup / Shutdown
Server Logs
Options File
PERFORMANCE
Dashboard
Performance Reports
Limit to 1000 rows
create table deal_with(
employee_id_FK int ,
constraint emmployee_FK foreign key (employee_id_FK) references employee (employee_id),
customer_ssn_FK int,
constraint customer_FK foreign key (customer_ssn_FK) references customer (customer_ssn)
);
Object Info
Session
No object selected


Final_project_DR_Hayam - Warning - not supported
Administration
Schemas
Query 1
SQL File 3*
SQL File 1*
SQL File 2*
SQL File 4*
SQL File 5*
MANAGEMENT
Server Status
Client Connections
Users and Privileges
Status and System Variables
Data Export
Data Import/Restore
INSTANCE
Startup / Shutdown
Server Logs
Options File
PERFORMANCE
Dashboard
Performance Reports
Limit to 1000 rows
create table phone(
phone nvarchar(11) primary key,
cus_ssn_FK int ,
constraint cus_FK foreign key (cus_ssn_FK) references customer (customer_ssn)
);
Object Info
Session
No object selected

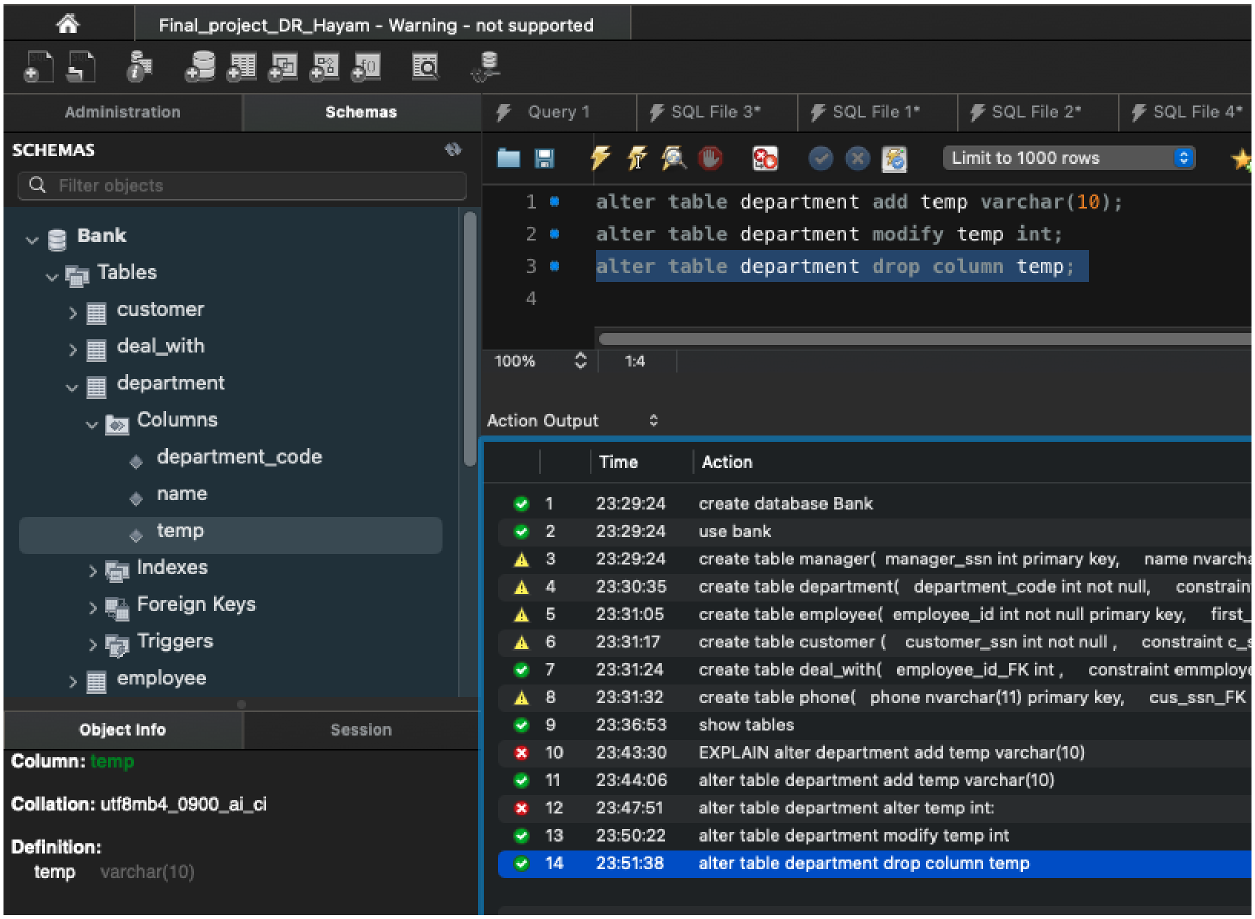
- **ALTER "ADD"**



MODIFY "ALTER"



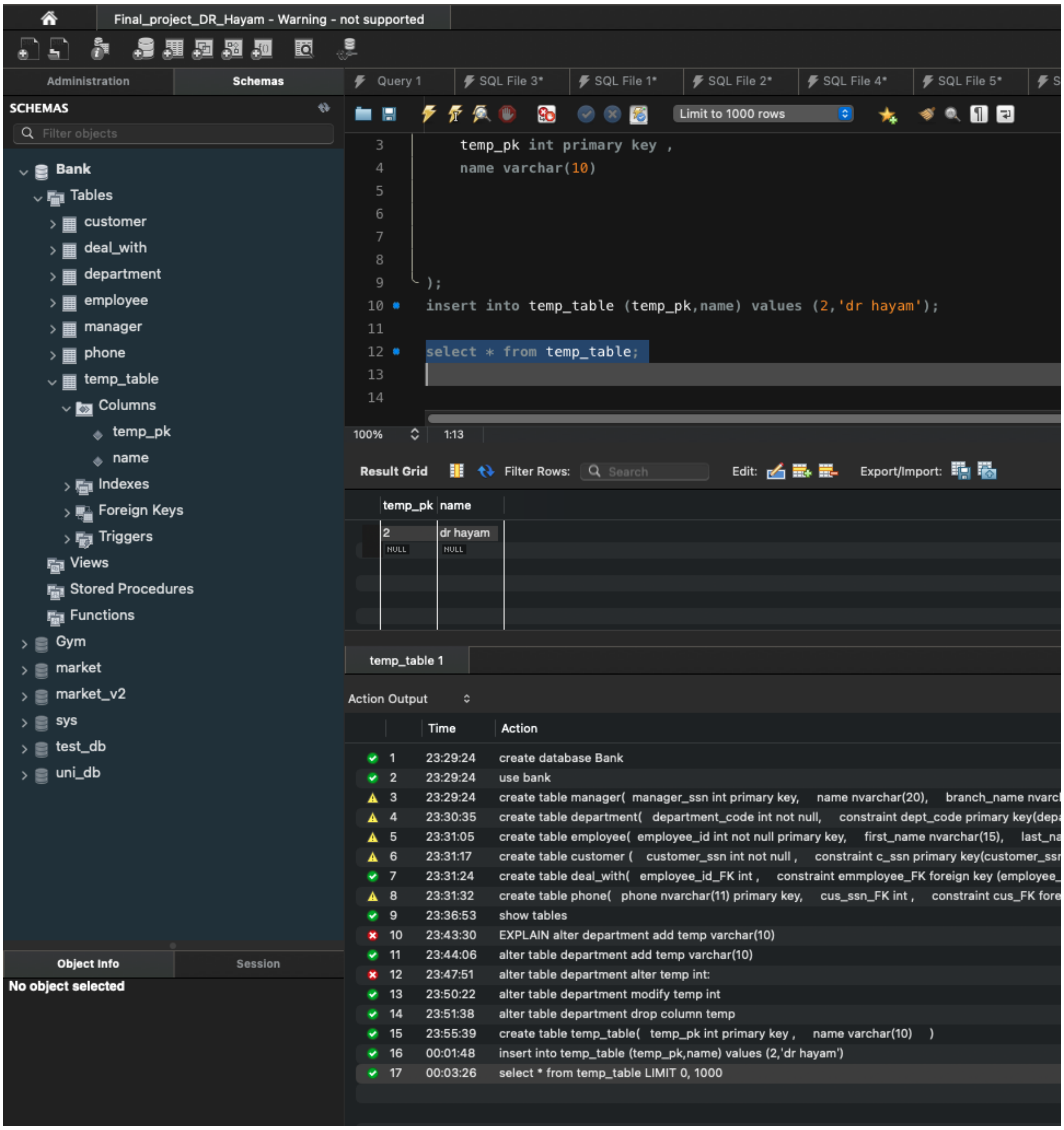
DROP



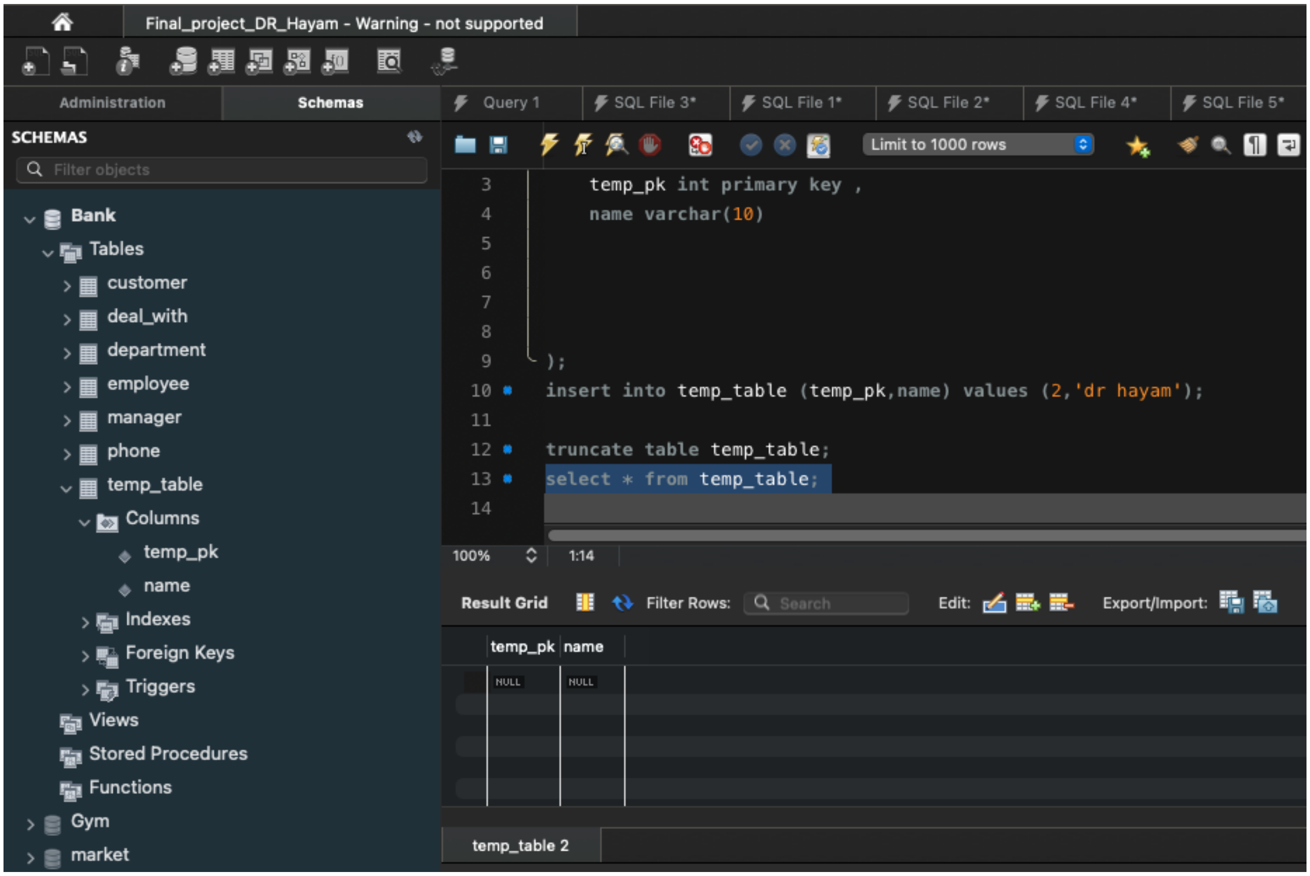
Final_project_DR_Hayam - Warning - not supported
Administration
Schemas
SCHEMAS
Bank
Tables
customer
deal_with
department
Columns
department_code
name
temp
Indexes
Foreign Keys
Triggers
employee
Query 1
SQL File 3*
SQL File 1*
SQL File 2*
SQL File 4*
Limit to 1000 rows
alter table department add temp varchar(10);
alter table department modify temp int;
alter table department drop column temp;
Action Output
Time
Action
1 23:29:24 create database Bank
2 23:29:24 use bank
9 23:36:53 show tables
10 23:43:30 EXPLAIN alter department add temp varchar(10)
11 23:44:06 alter table department add temp varchar(10)
12 23:47:51 alter table department alter temp int:
13 23:50:22 alter table department modify temp int
14 23:51:38 alter table department drop column temp
Object Info
Session
Column: temp
Collation: utf8mb4_0900_ai_ci
Definition:
temp varchar(10)

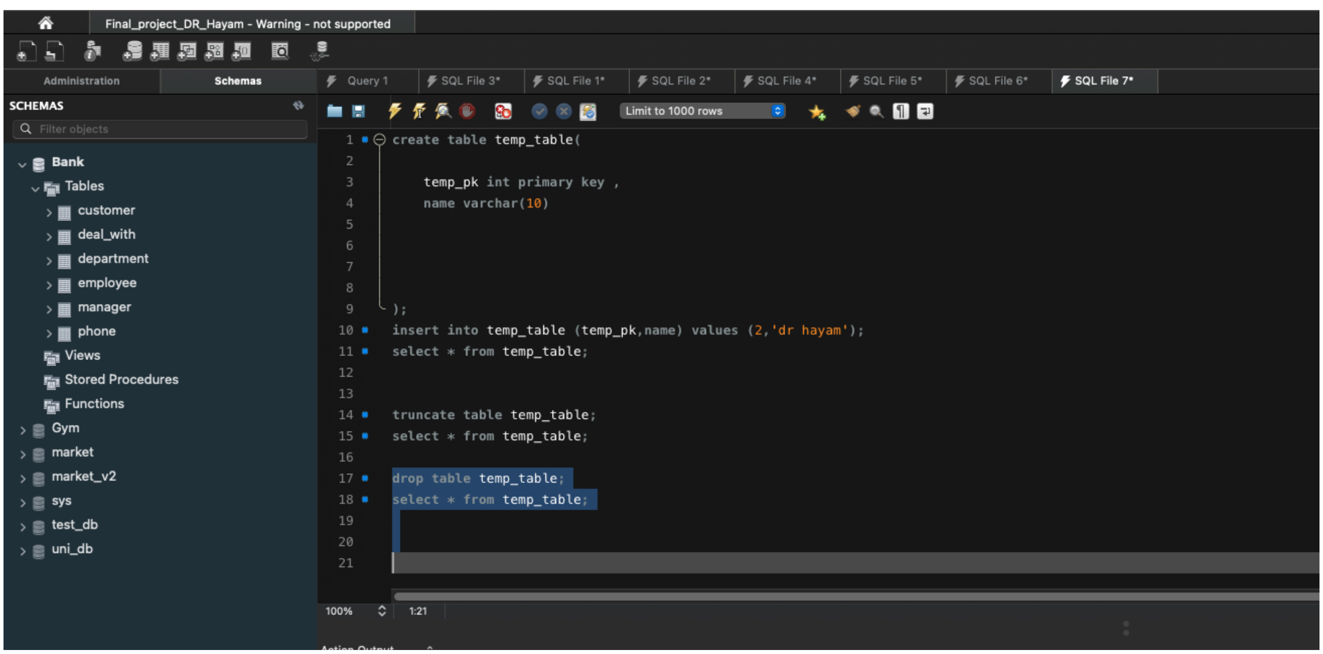
- # TRUNCATE & DELETE

BEFOOOOOOOOORE

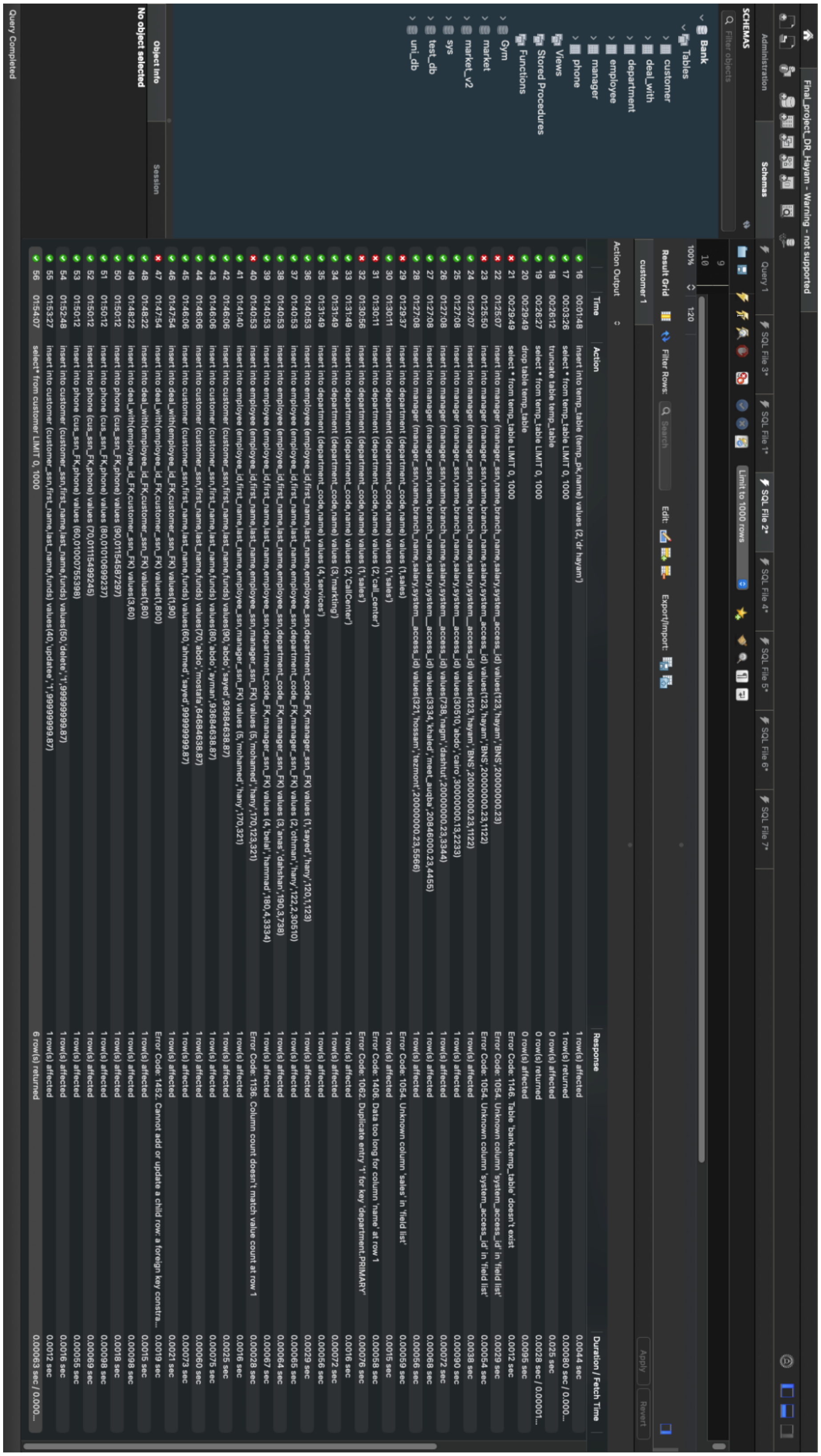


AFTEEEEEEEEEEER



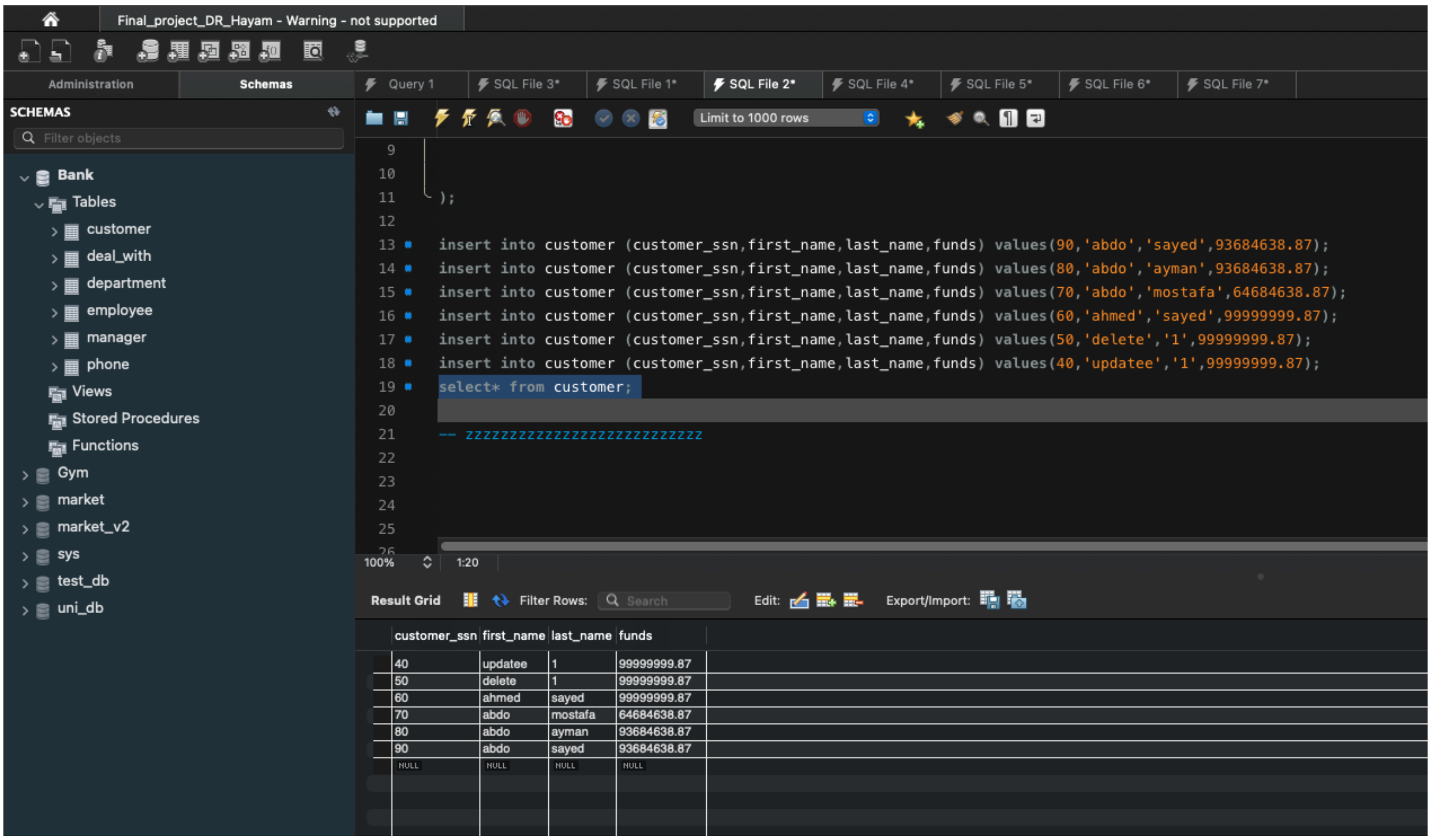


- **DML:- INSERT**

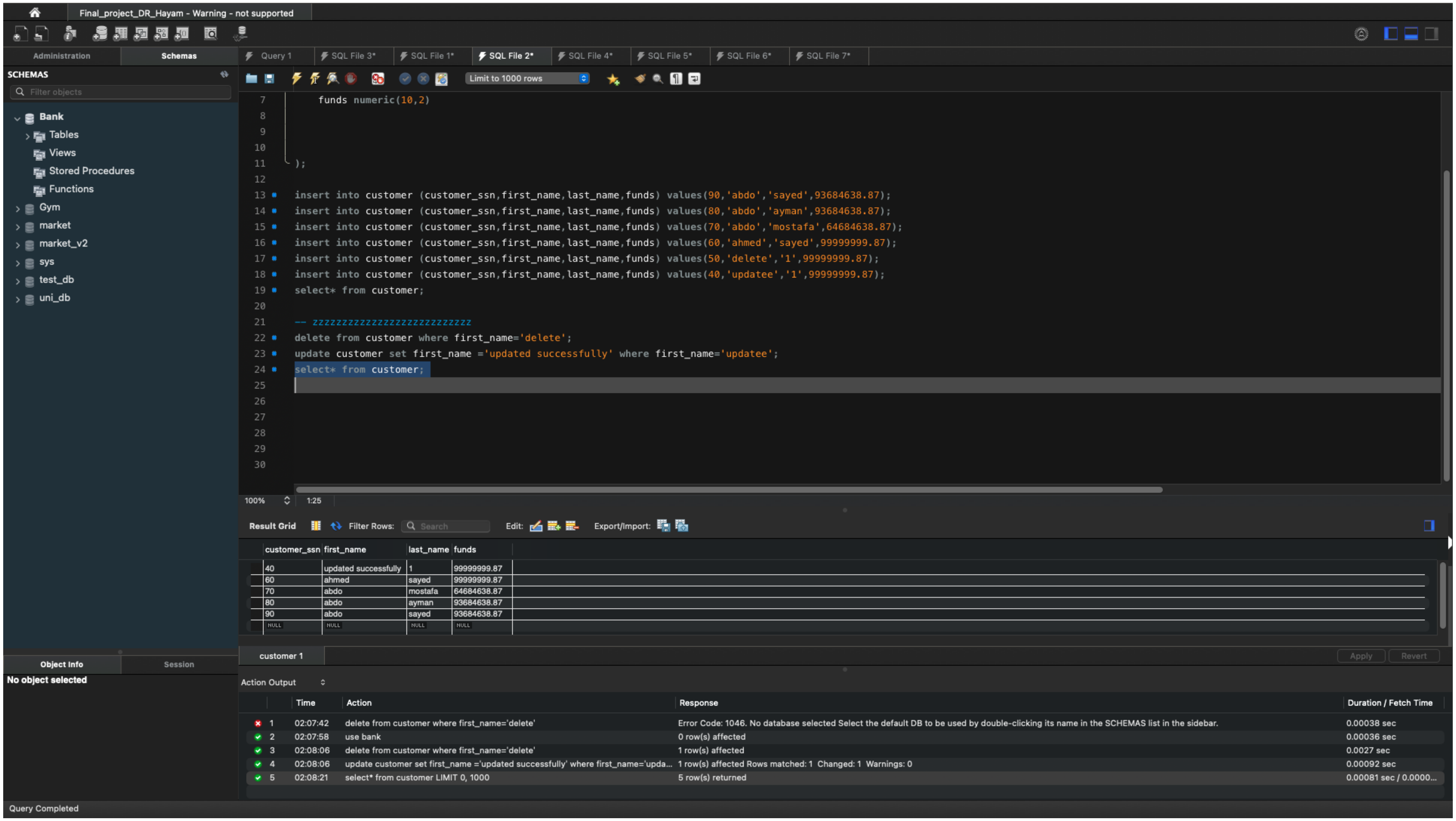


- **UPDATE**

BEFOOOOOOOOOOOORE

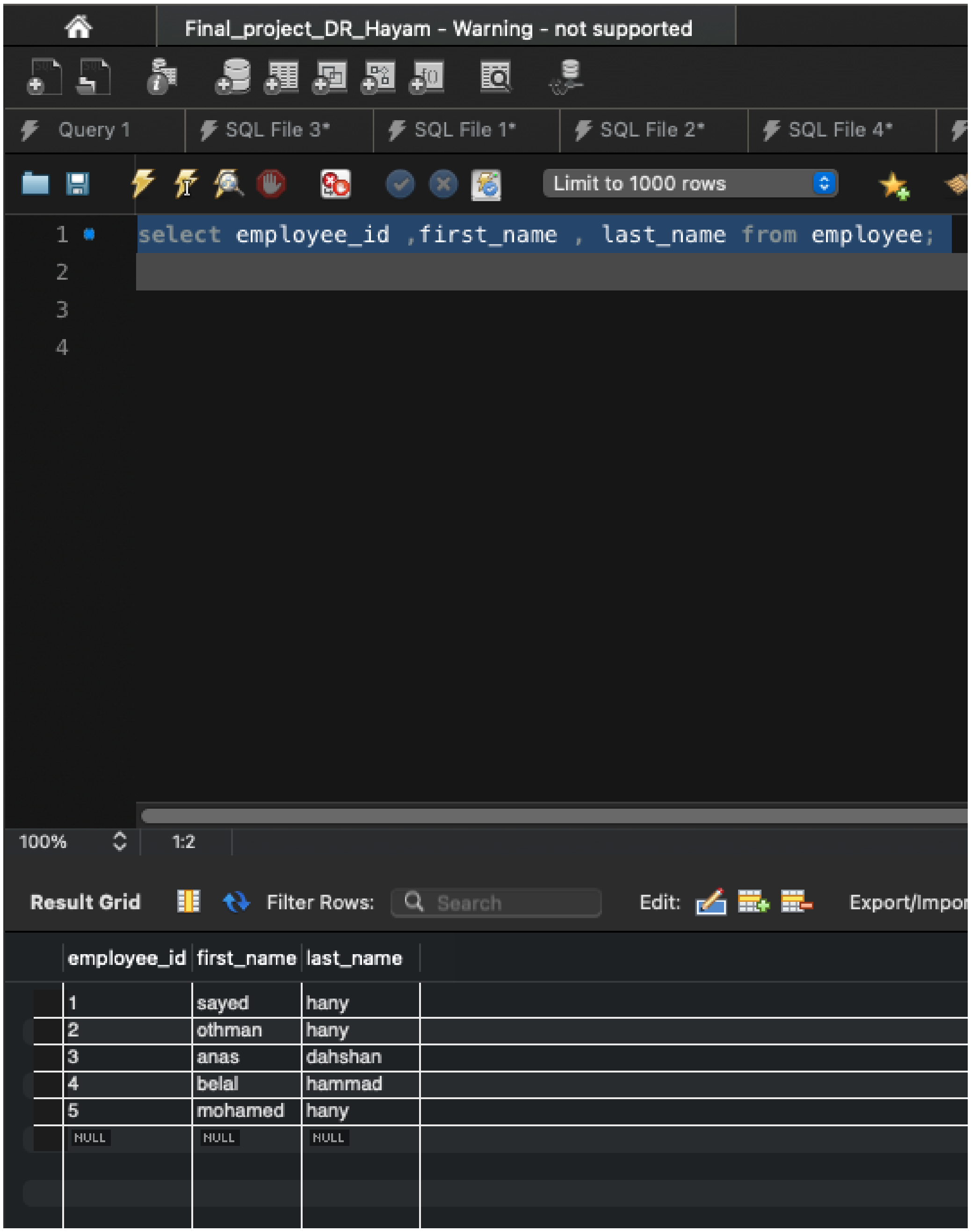


AFFFFFFFFFFTER

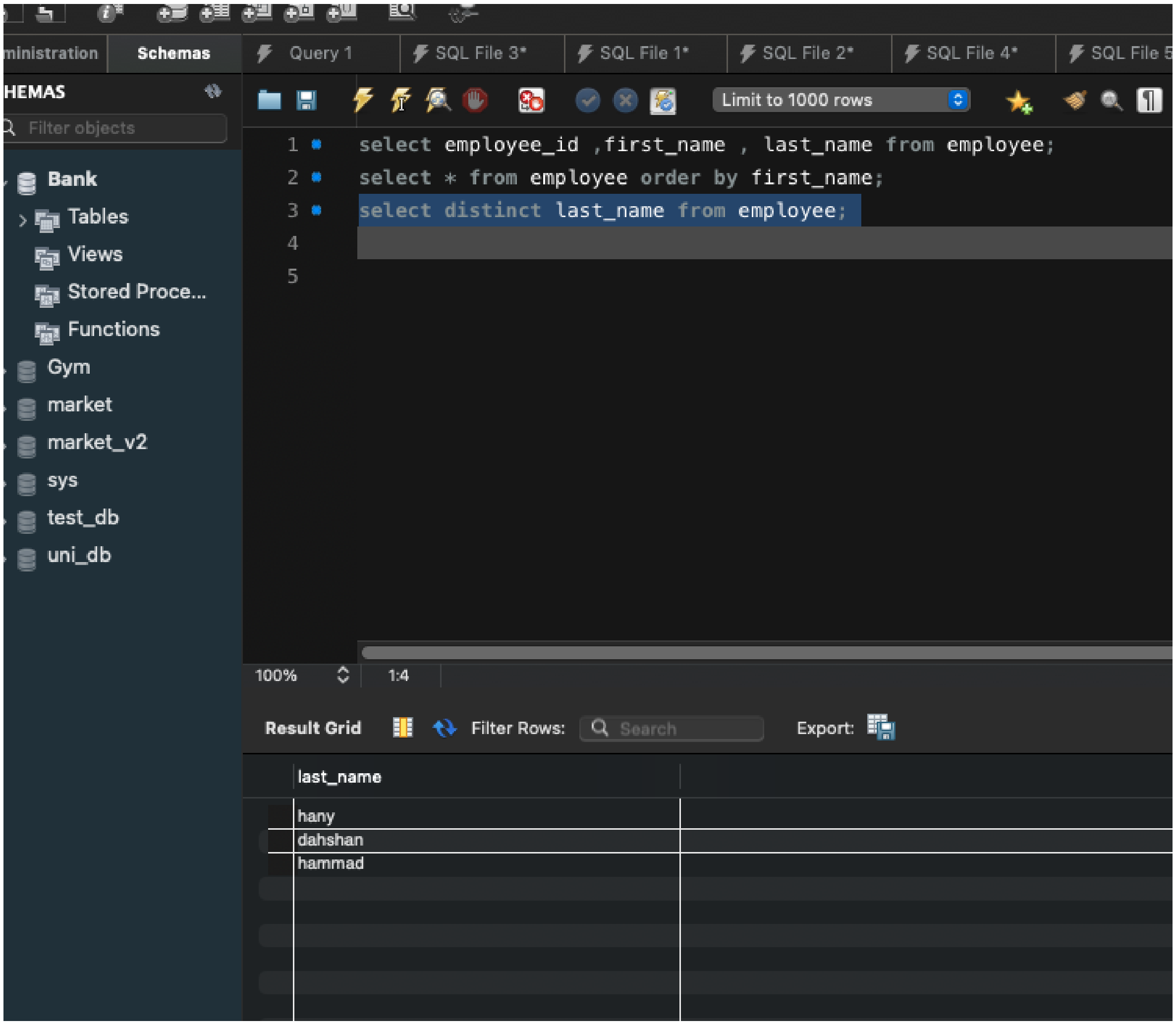

Final_project_DR_Hayam - Warning - not supported
Administration
Schemas
SCHEMAS
Bank
Tables
Views
Stored Procedures
Functions
Gym
market
market_v2
sys
test_db
uni_db
Query 1
SQL File 3*
SQL File 1*
SQL File 2*
SQL File 4*
SQL File 5*
SQL File 6*
SQL File 7*
Limit to 1000 rows
funds numeric(10,2)
);
insert into customer (customer_ssn,first_name,last_name,funds) values(90,'abdo','sayed',93684638.87);
insert into customer (customer_ssn,first_name,last_name,funds) values(80,'abdo','ayman',93684638.87);
insert into customer (customer_ssn,first_name,last_name,funds) values(70,'abdo','mostafa',64684638.87);
insert into customer (customer_ssn,first_name,last_name,funds) values(60,'ahmed','sayed',99999999.87);
insert into customer (customer_ssn,first_name,last_name,funds) values(50,'delete','1',99999999.87);
insert into customer (customer_ssn,first_name,last_name,funds) values(40,'updatee','1',99999999.87);
select* from customer;
-- zzzzzzzzzzzzzzzzzzzzzzzzz
delete from customer where first_name='delete';
update customer set first_name ='updated successfully' where first_name='updatee';
select* from customer;
Result Grid
Filter Rows:
Edit:
Export/Import:
customer_ssn first_name last_name funds
40 updated successfully 1 99999999.87
60 ahmed sayed 99999999.87
70 abdo mostafa 64684638.87
80 abdo ayman 93684638.87
90 abdo sayed 93684638.87
customer 1
Apply
Revert
Object Info
Session
No object selected
Action Output
Time Action Response Duration / Fetch Time
1 02:07:42 delete from customer where first_name='delete' Error Code: 1046. No database selected Select the default DB to be used by double-clicking its name in the SCHEMAS list in the sidebar. 0.00038 sec
2 02:07:58 use bank 0 row(s) affected 0.00036 sec
3 02:08:06 delete from customer where first_name='delete' 1 row(s) affected 0.0027 sec
4 02:08:06 update customer set first_name ='updated successfully' where first_name='upda... 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0 0.00092 sec
5 02:08:21 select* from customer LIMIT 0, 1000 5 row(s) returned 0.00081 sec / 0.0000...
Query Completed

SELECT

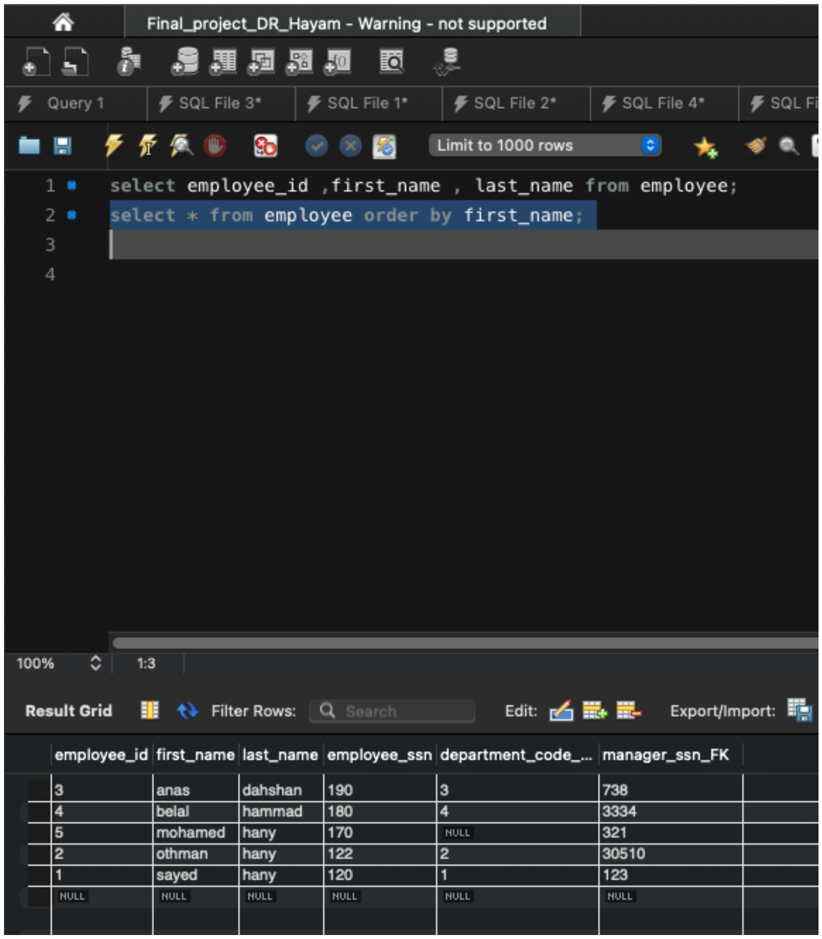
- ## BASIC



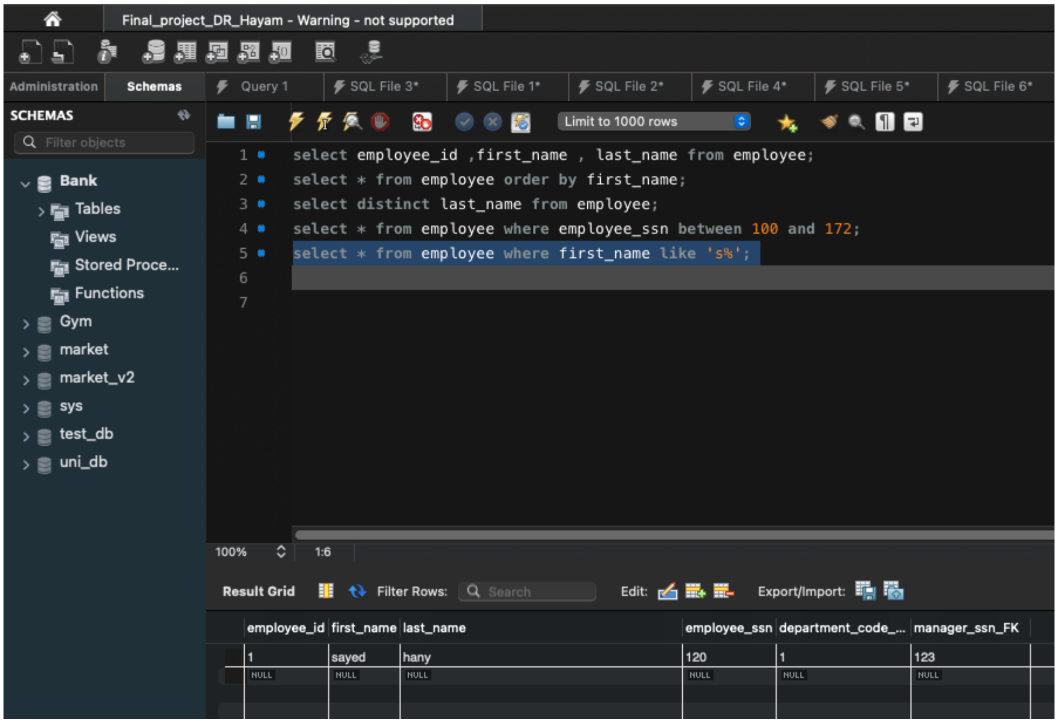
- ## DISTINCT

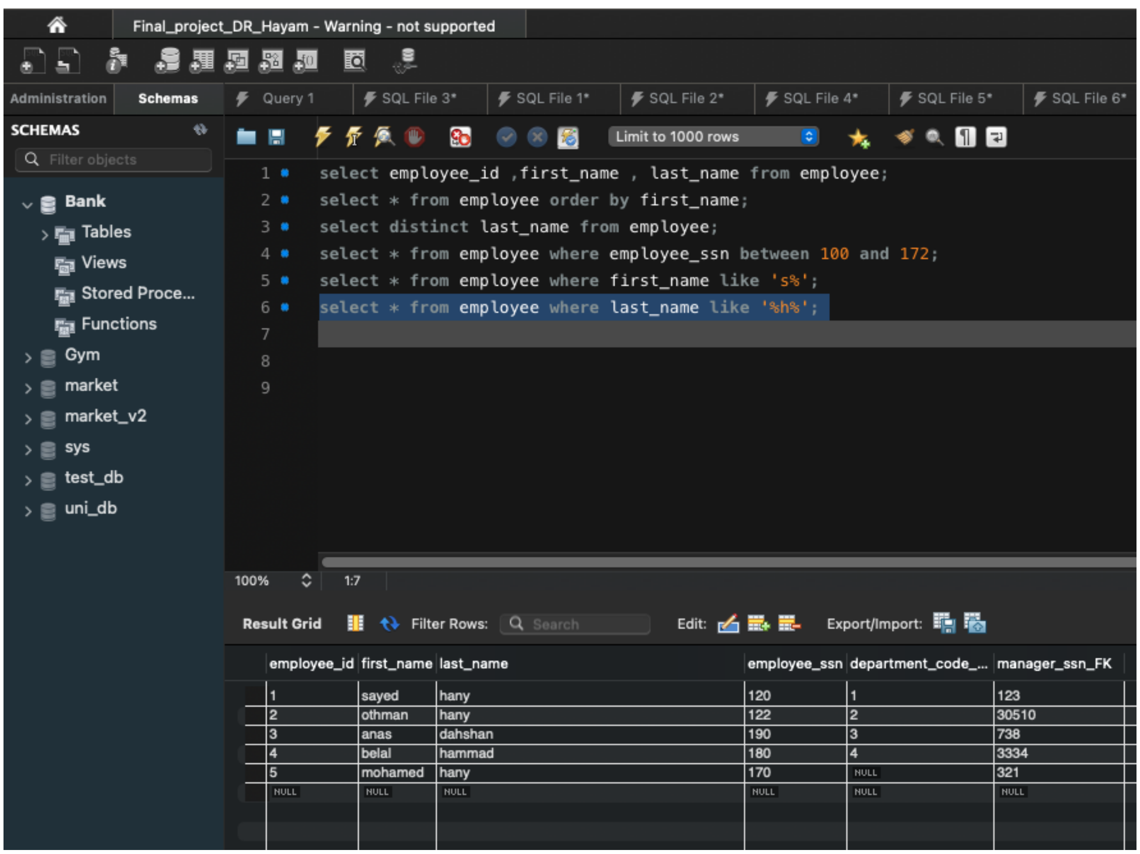

ministration
Schemas
Query 1
SQL File 3*
SQL File 1*
SQL File 2*
SQL File 4*
SQL File 5
HEMAS
Filter objects
Limit to 1000 rows
Bank
Tables
Views
Stored Proce...
Functions
Gym
market
market_v2
sys
test_db
uni_db
select employee_id ,first_name , last_name from employee;
select * from employee order by first_name;
select distinct last_name from employee;
100%
1:4
Result Grid
Filter Rows:
Search
Export:
last_name
hany
dahshan
hammad

- **ORDER BY**

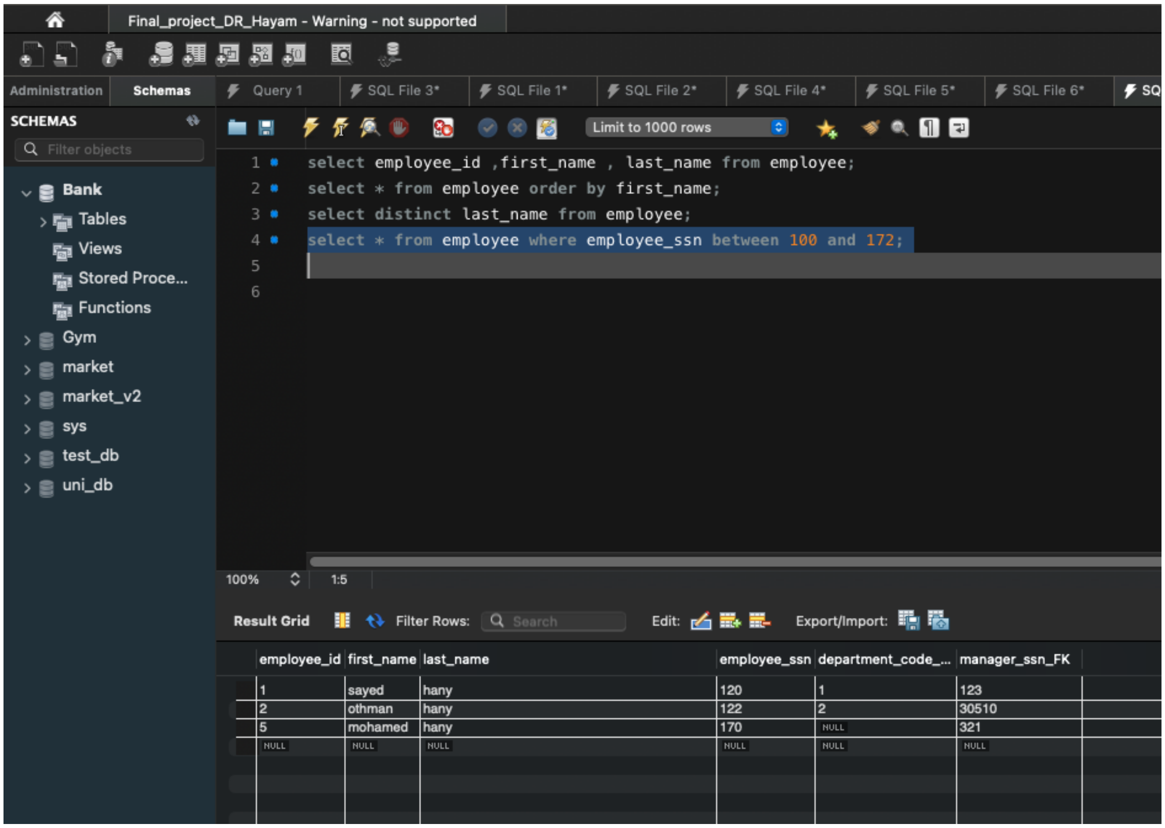

Final_project_DR_Hayam - Warning - not supported
Query 1
SQL File 3*
SQL File 1*
SQL File 2*
SQL File 4*
Limit to 1000 rows
select employee_id ,first_name , last_name from employee;
select * from employee order by first_name;
100%
1:3
Result Grid
Filter Rows:
Search
Edit:
Export/Import:
employee_id first_name last_name employee_ssn department_code_... manager_ssn_FK
3 anas dahshan 190 3 738
4 belal hammad 180 4 3334
5 mohamed hany 170 NULL 321
2 othman hany 122 2 30510
1 sayed hany 120 1 123
NULL NULL NULL NULL NULL NULL

- **WHERE LIKE**


Final_project_DR_Hayam - Warning - not supported
Administration
Schemas
Query 1
SQL File 3*
SQL File 1*
SQL File 2*
SQL File 4*
SQL File 5*
SQL File 6*
SCHEMAS
Filter objects
Bank
Tables
Views
Stored Proce...
Functions
Gym
market
market_v2
sys
test_db
uni_db
Limit to 1000 rows
select employee_id ,first_name , last_name from employee;
select * from employee order by first_name;
select distinct last_name from employee;
select * from employee where employee_ssn between 100 and 172;
select * from employee where first_name like 's%';
100%
1:6
Result Grid
Filter Rows:
Search
Edit:
Export/Import:
employee_id first_name last_name employee_ssn department_code_... manager_ssn_FK
1 sayed hany 120 1 123
NULL NULL NULL NULL NULL NULL

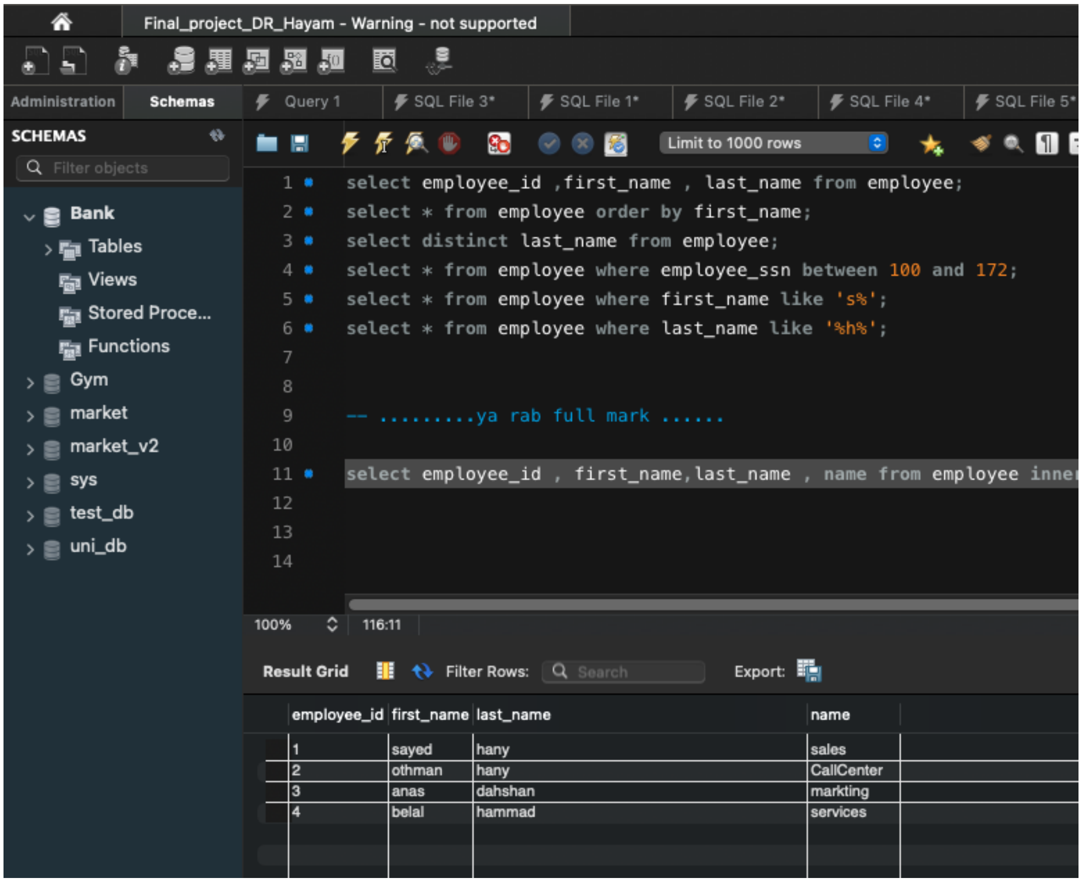

Final_project_DR_Hayam - Warning - not supported
Administration
Schemas
Query 1
SQL File 3*
SQL File 1*
SQL File 2*
SQL File 4*
SQL File 5*
SQL File 6*
SCHEMAS
Filter objects
Bank
Tables
Views
Stored Proce...
Functions
Gym
market
market_v2
sys
test_db
uni_db
Limit to 1000 rows
select employee_id ,first_name , last_name from employee;
select * from employee order by first_name;
select distinct last_name from employee;
select * from employee where employee_ssn between 100 and 172;
select * from employee where first_name like 's%';
select * from employee where last_name like '%h%';
100%
1:7
Result Grid
Filter Rows:
Search
Edit:
Export/Import:
employee_id first_name last_name employee_ssn department_code_... manager_ssn_FK
1 sayed hany 120 1 123
2 othman hany 122 2 30510
3 anas dahshan 190 3 738
4 belal hammad 180 4 3334
5 mohamed hany 170 NULL 321
NULL NULL NULL NULL NULL NULL

BETWEEN

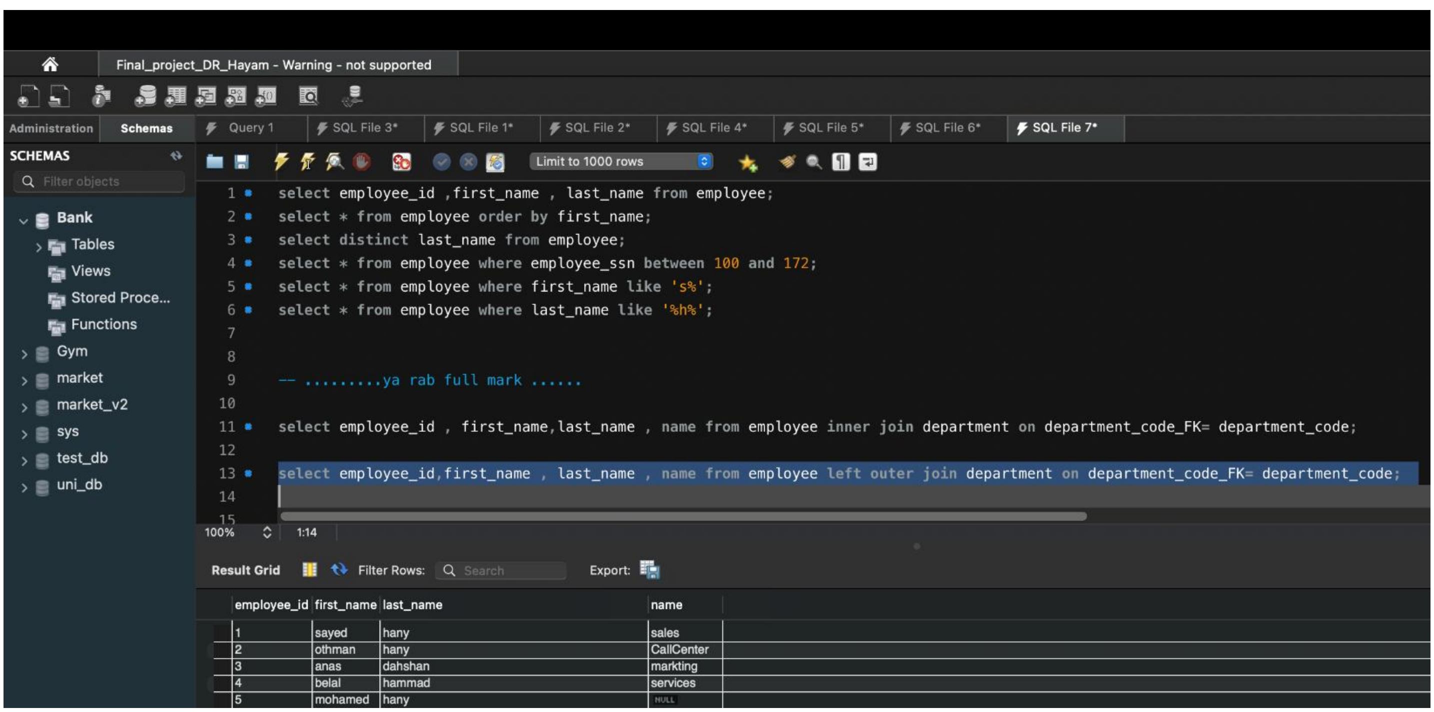


- **JOIN**

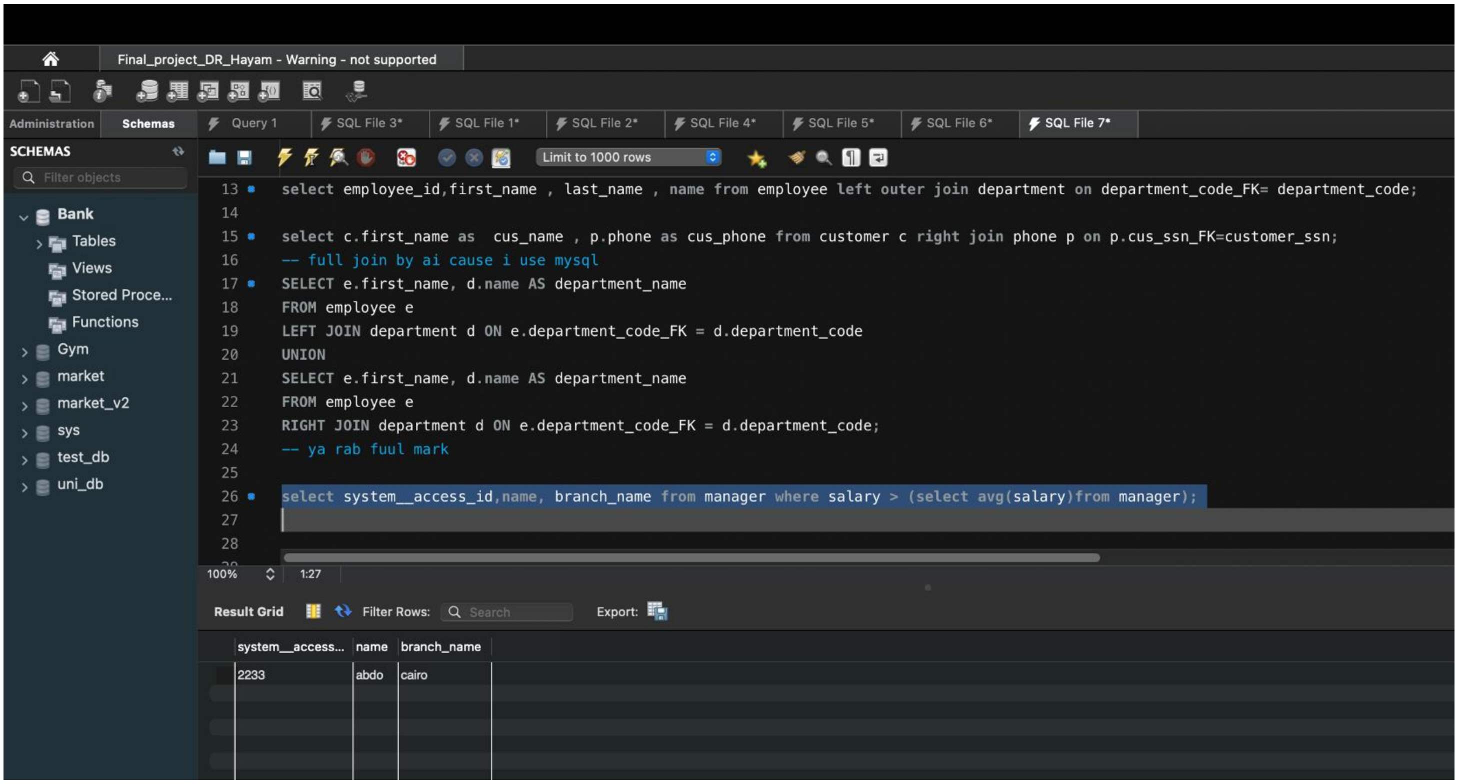
  INNER


Final_project_DR_Hayam - Warning - not supported
Administration
Schemas
Query 1
SQL File 3*
SQL File 1*
SQL File 2*
SQL File 4*
SQL File 5*
SCHEMAS
Filter objects
Limit to 1000 rows
Bank
Tables
Views
Stored Proce...
Functions
Gym
market
market_v2
sys
test_db
uni_db
select employee_id ,first_name , last_name from employee;
select * from employee order by first_name;
select distinct last_name from employee;
select * from employee where employee_ssn between 100 and 172;
select * from employee where first_name like 's%';
select * from employee where last_name like '%h%';
-- ....ya rab full mark ......
select employee_id , first_name,last_name , name from employee inne
100%
116:11
Result Grid
Filter Rows:
Search
Export:
employee_id first_name last_name name
1 sayed hany sales
2 othman hany CallCenter
3 anas dahshan markting
4 belal hammad services

LEFT


Final_project_DR_Hayam - Warning - not supported
Administration
Schemas
Query 1
SQL File 3*
SQL File 1*
SQL File 2*
SQL File 4*
SQL File 5*
SQL File 6*
SQL File 7*
SCHEMAS
Filter objects
Limit to 1000 rows
Bank
Tables
Views
Stored Proce...
Functions
Gym
market
market_v2
sys
test_db
uni_db
select employee_id ,first_name , last_name from employee;
select * from employee order by first_name;
select distinct last_name from employee;
select * from employee where employee_ssn between 100 and 172;
select * from employee where first_name like 's%';
select * from employee where last_name like '%h%';
-- ....ya rab full mark ......
select employee_id , first_name,last_name , name from employee inner join department on department_code_FK= department_code;
select employee_id,first_name , last_name , name from employee left outer join department on department_code_FK= department_code;
100%
1:14
Result Grid
Filter Rows:
Search
Export:
employee_id first_name last_name name
1 sayed hany sales
2 othman hany CallCenter
3 anas dahshan markting
4 belal hammad services
5 mohamed hany NULL

- **SUBQUERY & AGG FUNC**



- **VIEW**